

Error Analysis

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Precision and Recall

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Confusion Matrix

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Nearest Neighbors

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Explain the limit for Error analysis and controlled follow-up. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Error analysis and controlled follow-up. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Old errors are 5/10 on route A and 2/20 on B. New errors are 3/10 and 5/20. State total and group changes and a justified next step.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A model fails 6/10 backlit and 2/20 front-lit photos. Compute error rates and design a follow-up that tests lighting rather than leaf identity.

6. Solve the given case

Four errors occur among five backlit photos; two among twenty ordinary photos.

Compare rates and investigate.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Of eight errors, six are on blurry images and two on clear images. Can you conclude blurry images have a higher error rate without group totals?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 18 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Create the Plant Classifier error log linking case ID, actual, predicted, confidence, condition, likely cause, and next action.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A model fails 6/10 backlit and 2/20 front-lit photos. Compute error rates and design a follow-up that tests lighting rather than leaf identity.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Find the rates

Calculate the original error rates by lighting condition.

My reasoning and evidence

13. W2 Explain the limit

Does the difference prove dim light causes model errors?

My reasoning and evidence

14. W3 Track regressions

Summarize the changed results without hiding a regression.

My reasoning and evidence

15. W4 Apply independently

Old errors are 5/10 on route A and 2/20 on B. New errors are 3/10 and 5/20. State total and group changes and a justified next step.

My reasoning and evidence

16. W5 Repair a misleading audit

A team removes all disputed records only from the new model report. Why is the comparison invalid?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Error Analysis

Find an error cluster before choosing a controlled follow-up.

Candidate errors: bright-light photos1/8; dim-light photos3/6. Images and reference labels are fixed for this audit.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

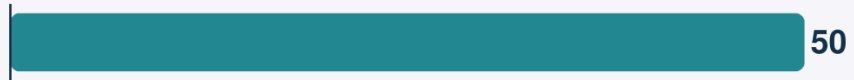
Errors per 100 photos in each lighting slice

percent

bright



dim



Bar length shows the supplied value; the marked baseline is zero.

Error Analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice errors

Bright: 12.5%; dim: 50%. Total errors 4/14, about 28.6%.



2. Inspect dim failures

Check blur, lighting and label uncertainty instead of assuming lighting caused every error.



3. Plan controlled pair

Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

CHANGE ONE THING

Dim errors fall to 1/6 in a later same-case replay.

Explain the changed result using the matching lesson.

What evidence would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Find an error cluster before choosing a controlled follow-up.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

1. Plan controlled pair
2. Compute slice errors
3. Inspect dim failures

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Plan controlled pair Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. Read the picture in order. Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Dim errors fall to $1/6$ in a later same-case replay. Which conclusion is supported by the supplied facts?

1. The replay's dim error rate is 16.7%; it is development evidence if those cases guided the revision.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The replay's dim error rate is 16.7%; it is development evidence if those cases guided the revision. Observed improvement on reused cases is not an untouched final generalisation test.

Repair a missing building block

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and prediction pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank training neighbours. Short practice: Why is the $k=1$ outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
